

# 3

## Discrete Random Variables and Probability Distributions

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## 3 Discrete Random Variables and Probability Distributions

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## 3.4 The Binomial Probability Distribution

# The Binomial Probability Distribution

There are many experiments that conform either exactly or approximately to the following list of requirements:

1. The experiment consists of **a sequence of  $n$  smaller experiments called *trials***, where  $n$  is fixed in advance of the experiment.
2. Each trial can result in one of the same **two possible outcomes** (dichotomous trials), which we generically denote by **success ( $S$ ) and failure ( $F$ )**.
3. **The trials are independent**, so that the outcome on any particular trial does not influence the outcome on any other trial.

# The Binomial Probability Distribution

4. The probability of success  $P(S)$  is constant from trial to trial; we denote this probability by  $p$ .

## Definition

An experiment for which Conditions 1–4 are satisfied is called a **binomial experiment**.

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## Example 27

The same coin is tossed successively and independently  $n$  times.

We arbitrarily use  $S$  to denote the outcome  $H$  (heads) and  $F$  to denote the outcome  $T$  (tails). Then this experiment satisfies Conditions 1–4.

Tossing a thumbtack  $n$  times, with  $S$  = point up and  $F$  = point down, also results in a binomial experiment.

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## Example 29

Sampling with replacement vs sampling without replacement.

The pool of participants consists of 50 individuals, of whom 35 are employed.

Suppose that 6 of these individuals are randomly selected one by one, each winning a prize.

Label the  $i$ th person selected (the  $i$ th trial) as a success  $S$  if he or she is employed and a failure  $F$  otherwise.

Calculate:  $P(S \text{ on first trial})$        $P(S \text{ on second trial})$

$P(S \text{ on sixth trial} \mid SSSSS)$        $P(S \text{ on sixth trial} \mid FFFFF)$

## Example 29

$$P(S \text{ on first trial}) = \frac{35}{50} = .70$$

$$\begin{aligned} P(S \text{ on second trial}) &= P(SS) + P(FS) \\ &= P(\text{second } S \mid \text{first } S)P(\text{first } S) \\ &\quad + P(\text{second } S \mid \text{first } F)P(\text{first } F) \\ &= \frac{34}{49} \cdot \frac{35}{50} + \frac{35}{49} \cdot \frac{15}{50} = \frac{35}{50} \left( \frac{34}{49} + \frac{15}{49} \right) = \frac{35}{50} = .70 \end{aligned}$$

$$P(S \text{ on sixth trial} \mid SSSSS) = 30/45 = .667$$

$$P(S \text{ on sixth trial} \mid FFFFF) = 35/45 = .778$$



# The Binomial Probability Distribution

## Rule

Consider **sampling without replacement** from a dichotomous population of size  $N$ .

If the **sample size** (number of trials)  $n$  is **at most 5% of the population size**, the experiment can be analyzed as though it were exactly a binomial experiment.

Consider sampling without replacement from a dichotomous population of size  $N$ . If the sample size (number of trials)  $n$  is at most 5% of the population size, the experiment can be analyzed as though it were a binomial experiment.



# The Binomial Random Variable and Distribution

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# The Binomial Random Variable and Distribution

In most binomial experiments, it is the total number of S's, rather than knowledge of exactly which trials yielded S's, that is of interest.

## Definition

The **binomial random variable  $X$**  associated with a binomial experiment consisting of  $n$  trials is defined as

**$X$  = the number of S's among the  $n$  trials**

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# The Binomial Random Variable and Distribution

Suppose, for example, that  $n = 3$ .

Then there are eight possible outcomes for the experiment:

$SSS$   $SSF$   $SFS$   $SFF$   $FSS$   $FSF$   $FFS$   $FFF$

From the definition of  $X$ ,  $X(SSF) = 2$ ,  $X(SFF) = 1$ , and so on. Possible values for  $X$  in an  $n$ -trial experiment are  $x = 0, 1, 2, \dots, n$ .

We will often write  $X \sim \text{Bin}(n, p)$  to indicate that  $X$  is a **binomial rv based on  $n$  trials with success probability  $p$ .**

# The Binomial Random Variable and Distribution

## Notation

Because the pmf of a binomial rv  $X$  depends on the **two parameters  $n$  and  $p$** , we denote the pmf by  **$b(x; n, p)$** .

## Example:

$n = 4$ , **calculate**  $p(3) = P(X = 3)$ ,

first calculate:

$$\begin{aligned} P(SSFS) &= P(S) \cdot P(S) \cdot P(F) \cdot P(S) \quad (\text{independent trials}) \\ &= p \cdot p \cdot (1 - p) \cdot p \quad (\text{constant } P(S)) \\ &= p^3 \cdot (1 - p) \end{aligned}$$

# The Binomial Random Variable and Distribution

second calculate:

$$b(3; 4, p) = P(FSSS) + P(SFSS) + P(SSFS) + P(SSSF) = 4p^3(1 - p)$$

**Table 3.1** Outcomes and Probabilities for a Binomial Experiment with Four Trials

| Outcome | $x$ | Probability    | Outcome | $x$ | Probability    |
|---------|-----|----------------|---------|-----|----------------|
| SSSS    | 4   | $p^4$          | FSSS    | 3   | $p^3(1 - p)$   |
| SSSF    | 3   | $p^3(1 - p)$   | FSSF    | 2   | $p^2(1 - p)^2$ |
| SSFS    | 3   | $p^3(1 - p)$   | FSFS    | 2   | $p^2(1 - p)^2$ |
| SSFF    | 2   | $p^2(1 - p)^2$ | FSFF    | 1   | $p(1 - p)^3$   |
| SFSS    | 3   | $p^3(1 - p)$   | FFSS    | 2   | $p^2(1 - p)^2$ |
| SFSF    | 2   | $p^2(1 - p)^2$ | FFSF    | 1   | $p(1 - p)^3$   |
| SFFS    | 2   | $p^2(1 - p)^2$ | FFFS    | 1   | $p(1 - p)^3$   |
| SFFF    | 1   | $p(1 - p)^3$   | FFFF    | 0   | $(1 - p)^4$    |

# The Binomial Random Variable and Distribution<sub>cont'd</sub>

the pmf of a binomial rv  $X$

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

## Example 31

Each of six randomly selected cola drinkers is given a glass containing cola  $S$  and one containing cola  $F$ . The glasses are identical in appearance except for a code on the bottom to identify the cola.

Suppose there is actually no tendency among cola drinkers to prefer one cola to the other.

Then  $p = P(\text{a selected individual prefers } S) = .5$ , so with  $X$  = the number among the six who prefer  $S$ ,  $X \sim \text{Bin}(6, .5)$ .

Thus

$$P(X = 3) = b(3; 6, .5) = \binom{6}{3} (.5)^3 (.5)^3 = 20(.5)^6 = .313$$



# Example 31

cont'd

The probability that **at least** three prefer S is

$$P(3 \leq X) = \sum_{x=3}^6 b(x; 6, .5) = \sum_{x=3}^6 \binom{6}{x} (.5)^x (.5)^{6-x} = .656$$

and the probability that **at most** one prefers S is

$$P(X \leq 1) = \sum_{x=0}^1 b(x; 6, .5) = .109$$

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# Using Binomial Tables

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# Using Binomial Tables

Even for a relatively small value of  $n$ , the computation of binomial probabilities can be tedious.

Appendix Table A.1 tabulates the cdf  $F(x) = P(X \leq x)$  for  $n = 5, 10, 15, 20, 25$  in combination with selected values of  $p$ .

Various other probabilities can then be calculated using the proposition on cdf's.

A table entry of 0 signifies only that the probability is 0 to three significant digits since all table entries are actually positive.

# Using Binomial Tables

## Notation

For  $X \sim \text{Bin}(n, p)$ , the cdf will be denoted by

$$B(x; n, p) = P(X \leq x) = \sum_{y=0}^x b(y; n, p) \quad x = 0, 1, \dots, n$$

For  $X \sim \text{Bin}(n, p)$ , the cdf will be denoted by

$$B(x; n, p) = P(X \leq x) = \sum_{y=0}^x b(y; n, p) \quad x = 0, 1, \dots, n$$

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# Using Binomial Tables

$$B(x; n, p) = \sum_{y=0}^x b(y; n, p)$$

**Table A.1** Cumulative Binomial Probabilities

a.  $n = 5$

|          |   | <i>p</i> |       |       |       |      |      |      |      |      |
|----------|---|----------|-------|-------|-------|------|------|------|------|------|
|          |   | 0.01     | 0.05  | 0.10  | 0.20  | 0.25 | 0.30 | 0.40 | 0.50 | 0.60 |
| <i>x</i> | 0 | .951     | .774  | .590  | .328  | .237 | .168 | .078 | .031 | .010 |
|          | 1 | .999     | .977  | .919  | .737  | .633 | .528 | .337 | .188 | .087 |
|          | 2 | 1.000    | .999  | .991  | .942  | .896 | .837 | .683 | .500 | .317 |
|          | 3 | 1.000    | 1.000 | 1.000 | .993  | .984 | .969 | .913 | .812 | .663 |
|          | 4 | 1.000    | 1.000 | 1.000 | 1.000 | .999 | .998 | .990 | .969 | .922 |



# The Mean and Variance of $X$

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# The Mean and Variance of $X$

For  $n = 1$ , the binomial distribution becomes the Bernoulli distribution.

The mean value of a Bernoulli variable is  $\mu = p$ , so the expected number of  $S$ 's on any single trial is  $p$ .

Since a binomial experiment consists of  $n$  trials, intuition suggests that for  $X \sim \text{Bin}(n, p)$ ,  $E(X) = np$ , the product of the number of trials and the probability of success on a single trial.

The expression for  $V(X)$  is not so intuitive.

# The Mean and Variance of $X$

## Proposition

If  $X \sim \text{Bin}(n, p)$ , then  $E(X) = np$ ,  $V(X) = np(1 - p) = npq$ , and  $\sigma_X = \sqrt{npq}$  (where  $q = 1 - p$ ).

If  $X \sim \text{Bin}(n, p)$ , then

$$E(X) = np, \quad V(X) = np(1 - p) = npq,$$

$$\sigma_X = \sqrt{npq}$$

(where  $q = 1 - p$ ).

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## Example 34

If 75% of all purchases at a certain store are made with a credit card and  $X$  is the number among ten randomly selected purchases made with a credit card, then  $X \sim \text{Bin}(10, .75)$ .

Thus  $E(X) = np = (10)(.75) = 7.5$ ,

$$V(X) = npq = 10(.75)(.25)$$

$$= 1.875,$$

$$\text{and } \sigma = \sqrt{1.875}$$

$$= 1.37.$$

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## Example 34

cont'd

Again, even though  $X$  can take on only integer values,  $E(X)$  need not be an integer.

If we perform a large number of independent binomial experiments, each with  $n = 10$  trials and  $p = .75$ , then the average number of  $S$ 's per experiment will be close to 7.5.

The probability that  $X$  is within 1 standard deviation of its mean value is

$$\begin{aligned} P(7.5 - 1.37 \leq X \leq 7.5 + 1.37) &= P(6.13 \leq X \leq 8.87) \\ &= P(X = 7 \text{ or } 8) \\ &= .532. \end{aligned}$$